AEGIS-TITAN SOURCE REVIEW REPORT

Engineering Evidence and Review Limitations
Review date: 06 November 2026
Reviewed source reference: 0be0ee4c57cd1e1a98527f4ae5b3a3efd6a61e94
Repository visibility: Private - access on request
Review type: AI-assisted inspection of selected source files

1. EXECUTIVE SUMMARY

The inspected source includes an event-driven core, network and host-integrity components, a digital-forensics analyzer, Linux enforcement code, and a persistent SQLite evidence ledger using SHA-256 hash chaining.
This is not an independent security audit, penetration test, or certification.

2. REVIEW SCOPE

Files inspected include core/contracts.py, core/event_bus.py, core/orchestrator.py, core/forensic_ledger.py, core/manual_registry.py, modules/network_intel/network_sentinel.py, modules/host_integrity/fim_sentinel.py, modules/dfir_core/digital_forensics.py, modules/active_defense/firewall_enforcer.py, and tests/test_defensive_engines.py.
Full repository file count and total lines of code were not established.

3. TECHNOLOGY EVIDENCE

Python 3.10+ (stated in contracts), Rich, asyncio, threading, sqlite3, hashlib, socket, struct, Pillow (PIL imports), and Scapy in wireless verification scripts.
No complete dependency manifest or pinned package-version inventory was verified.

4. VERIFIED SOURCE COMPONENTS

Event bus and contracts; network packet inspection; host file-integrity monitoring; digital forensics and artifact analysis; Linux firewall enforcement and file quarantine; persistent SQLite forensic ledger with SHA-256 block-chain verification.

5. TEST SOURCE REVIEW

tests/test_defensive_engines.py contains seven test_* methods and asserts that
the manual registry contains 22 entries. The test suite was not executed during
this review. No pass/fail result or coverage percentage is claimed.
Additional phase-verification scripts exist; their execution was not reproduced.

6. ENGINEERING DECISIONS AND TRADE-OFFS
Async event dispatch supports multiple consumers but increases concurrency complexity.
SQLite keeps evidence storage local and self-contained; distributed use needs more design.
Canonical JSON plus SHA-256 helps detect changes on re-verification; it is not a
replacement for digital signatures or immutable external storage.
Thread-to-async bridging integrates blocking workers but adds lifecycle complexity.
Linux iptables enforcement provides native control but reduces cross-platform portability.

7. FINDING
The inspected Event Bus _safe_dispatch catches handler exceptions and currently
uses pass. This isolates subscriber failures but does not provide visible logging
in that method. Structured exception logging and failure metrics are recommended.

8. ROADMAP
Execute tests in a controlled environment; add structured event logging; inventory
and pin dependencies; benchmark precision, recall and false positives; obtain an
independent security review before making certification claims.

9. EXPLICIT LIMITATIONS
Not verified: total file count, total LOC, full dependency versions, test execution,
coverage, benchmark accuracy, false-positive rate, production readiness, or
independent security certification.

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Method: AI-assisted source inspection of selected files at the stated source ref.